JAMES NICHOLAS PATTERSON

Data Scientist

Email: james.patterson@gmail.com

Phone: +44 7891 234567

LinkedIn: linkedin.com/in/jamespatterson92

GitHub: github.com/jnpatterson

PROFESSIONAL SUMMARY

I'm a data scientist with four years of experience building predictive models and analytics solutions for fintech and healthcare sectors. I enjoy the problem-solving side of data work, particularly when it involves messy real-world datasets. I've spent most of my career moving between Python and SQL, occasionally venturing into Scala when the data gets too big. I'm based in Cambridge and interested in roles where I can contribute to both model development and the infrastructure that supports it.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Scala, R

Machine Learning: Scikit-learn, XGBoost, TensorFlow, PyTorch, Statsmodels

Data Engineering: Apache Spark, Hadoop, Pandas, NumPy

Cloud Platforms: AWS (S3, EC2, SageMaker), Google Cloud Platform

DevOps Tools: Kubernetes, Terraform, ArgoCD, Docker

Databases: PostgreSQL, MySQL, MongoDB, Cassandra

Visualisation: Matplotlib, Seaborn, Plotly, Tableau

Version Control: Git, GitHub, GitLab

WORK EXPERIENCE

Senior Data Scientist

FinMetrics Solutions

finmetrics.io

Cambridge, United Kingdom

March 2023 - Present

Developed and deployed a fraud detection model using gradient boosting that reduced false positives by 35% and saved the company approximately GBP 2.1 million annually in chargebacks

Built an automated data pipeline in Python that processes transaction data from multiple banking partners, handling approximately 500 million records per month with 99.2% uptime

Created a feature engineering framework that standardised how the team approaches model input preparation, reducing time spent on manual data processing by 40%

Mentored two junior data scientists on statistical testing methods and established code review practices that improved model reproducibility across the team

Data Scientist

HeartTech Analytics

hearttech.net

Cambridge, United Kingdom

July 2021 - February 2023

Constructed predictive models for patient readmission risk using historical clinical data, achieving an AUC of 0.89 which outperformed the existing rule-based system by 12 percentage points

Implemented an end-to-end machine learning pipeline using Spark that processed electronic health records from seven NHS trusts, serving predictions to a web application used by 50 clinicians daily

Designed and conducted A/B tests on model outputs to measure clinical impact, working closely with hospital staff to validate that model predictions aligned with real-world patient outcomes

Led the migration of legacy R scripts to Python, converting three years of analytical work and enabling the team to integrate models with modern data infrastructure

Data Analyst

Cleardata Insights Ltd

www.cleardatainsights.com

London, United Kingdom

June 2019 - June 2021

Analysed customer behaviour data for retail clients using SQL and Python to identify purchasing patterns, resulting in three clients implementing targeted marketing campaigns that increased repeat purchases by 18-22%

Built dashboards in Tableau that tracked key performance indicators for twenty client accounts, automating manual reporting processes that previously took eight hours per week

Performed statistical analysis on A/B test results across multiple e-commerce platforms, providing actionable recommendations to product teams on feature releases

Supported the integration of new data sources by writing SQL transformations and validating data quality, ensuring 98% accuracy in the customer analytics warehouse

EDUCATION

MSc Data Science

University of Edinburgh

Edinburgh, United Kingdom

2017 - 2019

BSc Mathematics with Statistics

University of Cambridge

Cambridge, United Kingdom

2013 - 2017

PUBLIC REPOSITORIES

github.com/jnpatterson/fraud-detection-benchmark

Comparative analysis of tree-based and neural network approaches for financial fraud detection using public datasets

github.com/jnpatterson/time-series-health-forecasting

LSTM and ARIMA implementations for predicting patient deterioration from vital signs data

github.com/jnpatterson/feature-engineering-toolkit

Reusable Python library for automated feature generation from time-series and tabular data

github.com/jnpatterson/statistical-testing-guide

Documentation and worked examples of statistical tests commonly used in A/B testing and experimentation